

2(a)	<i>if only one molecule described, statement of difference must be in context of glycogen</i> any three from: <i>glycogen</i> branched (v cellulose unbranched) ; monomer is / made of, <u>alpha-</u> / <u>α-</u>, glucose (v cellulose monomer is β-glucose) ; (1,4- and) <u>1, 6-. glycosidic bond(s)</u> (v cellulose 1,4-glycosidic bond(s)) ; alternate, monomers / glucose residues / 1,4-(glycosidic) bonds, are not, rotated 180° / inverted / flipped / AW ; ora	3
2(b)	any three from: 1 glycogen is a <u>store</u> of, energy / glucose ; R produces energy 2 <i>idea that</i> many, branching points / 'ends' / terminals, for easy / faster, addition / release, of glucose ; 3 compact so stores much, glucose / energy (in cells / small space) ; 4 insoluble ; 5 does not, lower / change, water potential (of cell) ; A no osmotic effect 6 AVP ; e.g. prevents loss of glucose from cell (chemically) inactive so does not take part in cell metabolism / AW	3

Question	Answer	Marks
2(c)	any three from: 1 unbranched (polymer) / straight chain / linear ; 2 cellulose molecules are arranged in parallel ; 3 <i>idea that</i> large number of -OH groups (projecting in all directions) ; 4 allows, hydrogen bonds / H-bonds, to form with water or makes molecule hydrophilic ; 5 many hydrogen bonds between cellulose molecules / AW ; 6 cellulose molecules form <u>microfibrils</u> ; I 'fibrils' / fibres 7 hydrogen bonds give high tensile strength to microfibrils A cellulose fibres / cell wall or hydrogen bonds help cell walls to resist, turgor pressure / AW ;	3

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